AI Translator Guide

Converting raw country livestock data into the input template

CIAT / CGIAR Alliance, ClimateActionNetZero

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This page describes what the in-app AI translator does, how to use it for a single inventory, and how access is granted. It complements the user guide and the methodology document with a step-by-step walk-through of the translator.

1 What the AI translator is for

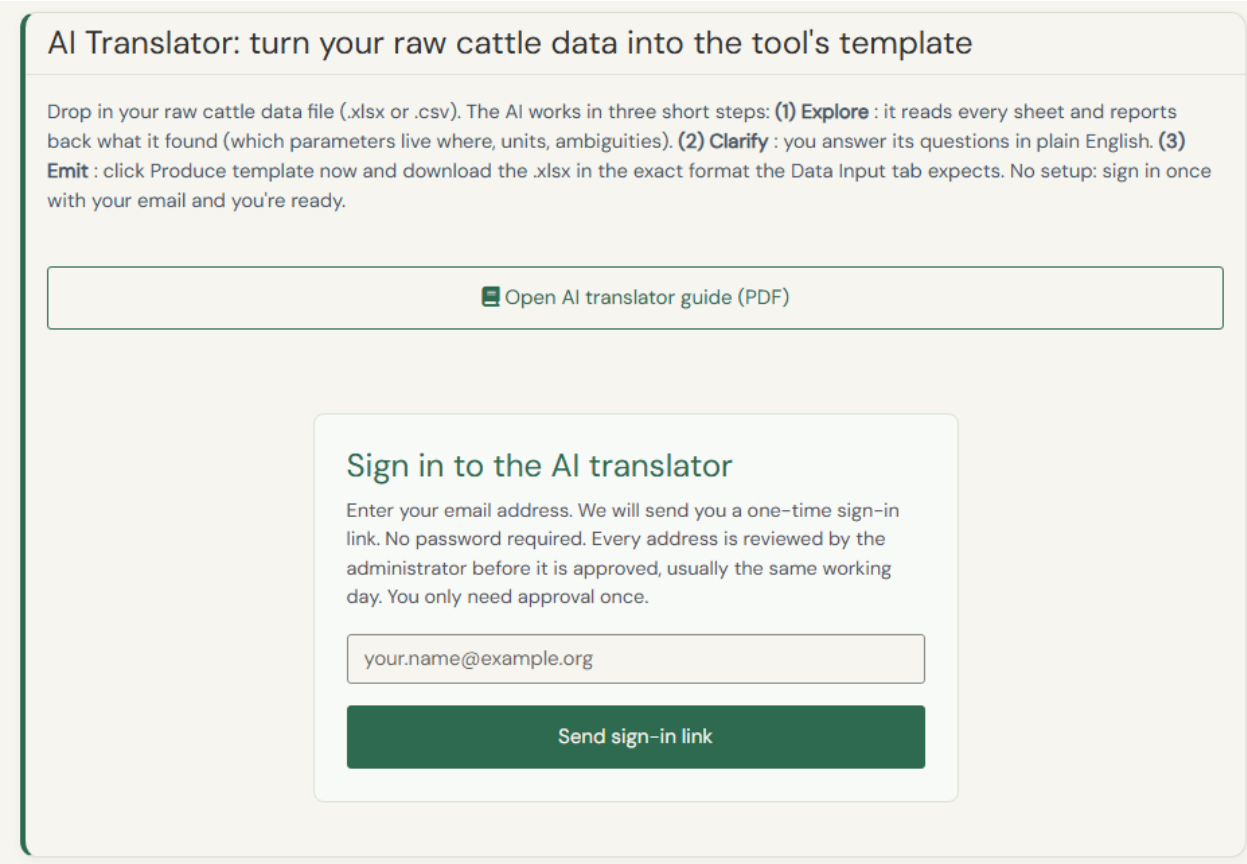
The Cattle Uncertainty App expects its data in a specific Excel template, so that it can run the uncertainty analysis. Real national inventory data rarely arrives in that shape. Column names differ. Units differ. Sub-category labels reflect local practice (“Local breed adult cows”, “draught oxen”).

The AI translator converts raw inventory data, in whatever shape it arrives, into the exact template the tool needs.

2 Step by step

2.1 1. Accessing the AI translator

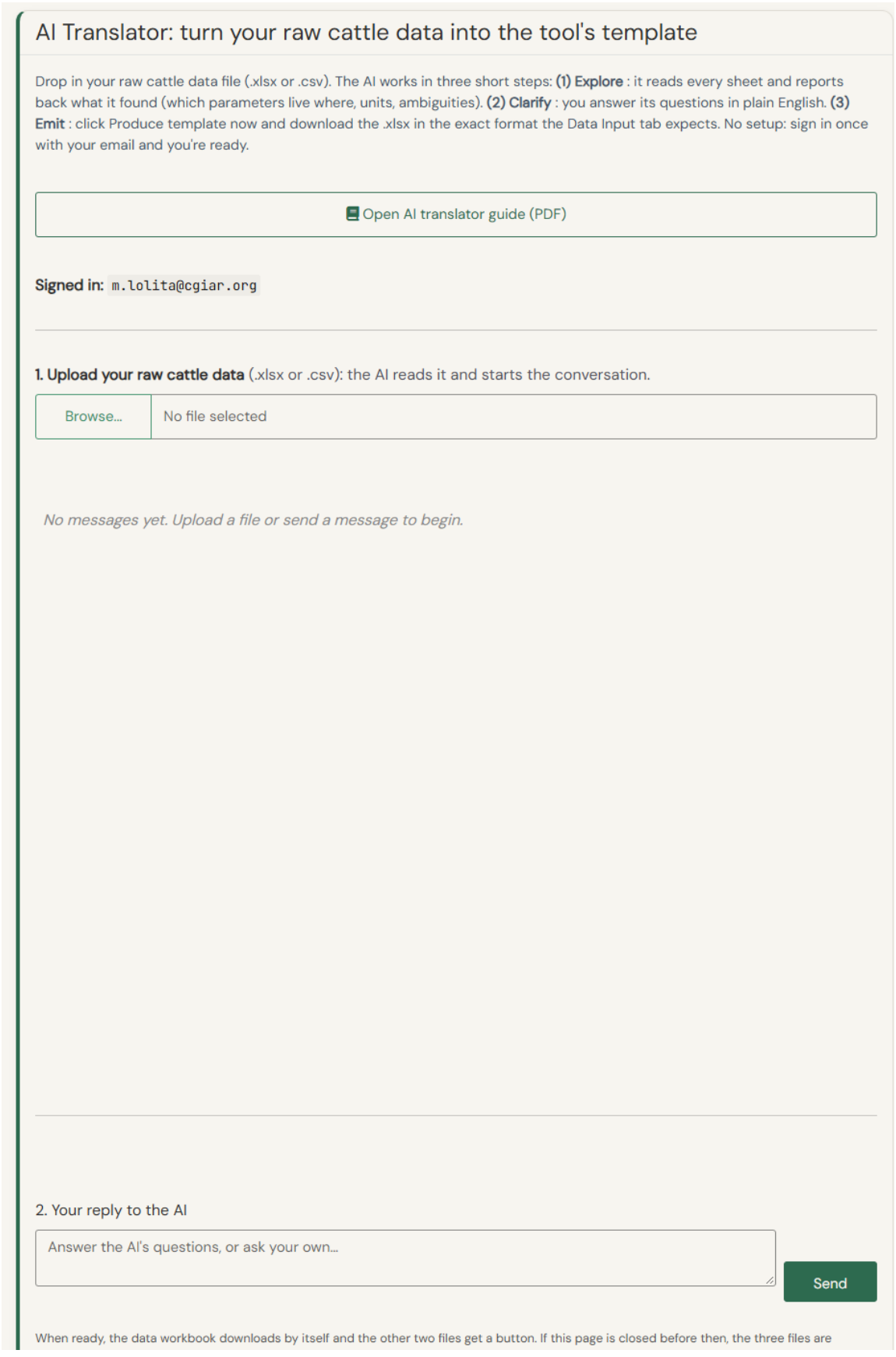
In the Resources tab of the app, the first thing you see is that you have to sign in to use the translator.



The translator is linked to an AI service and costs money on every run, so we control access. You ask for access with your email address. We review the request and accept your email.

1. Type your email address and click **Send sign-in link**. A confirmation email arrives with a link.
2. Click the link. The first time, the app tells you that your request has been sent to us for approval.
3. When your email is accepted, you receive a second email saying so, with a link to the app.
4. Click that link, or ask for a new sign-in link on the Resources tab. You are redirected to the app with access to the AI translator.

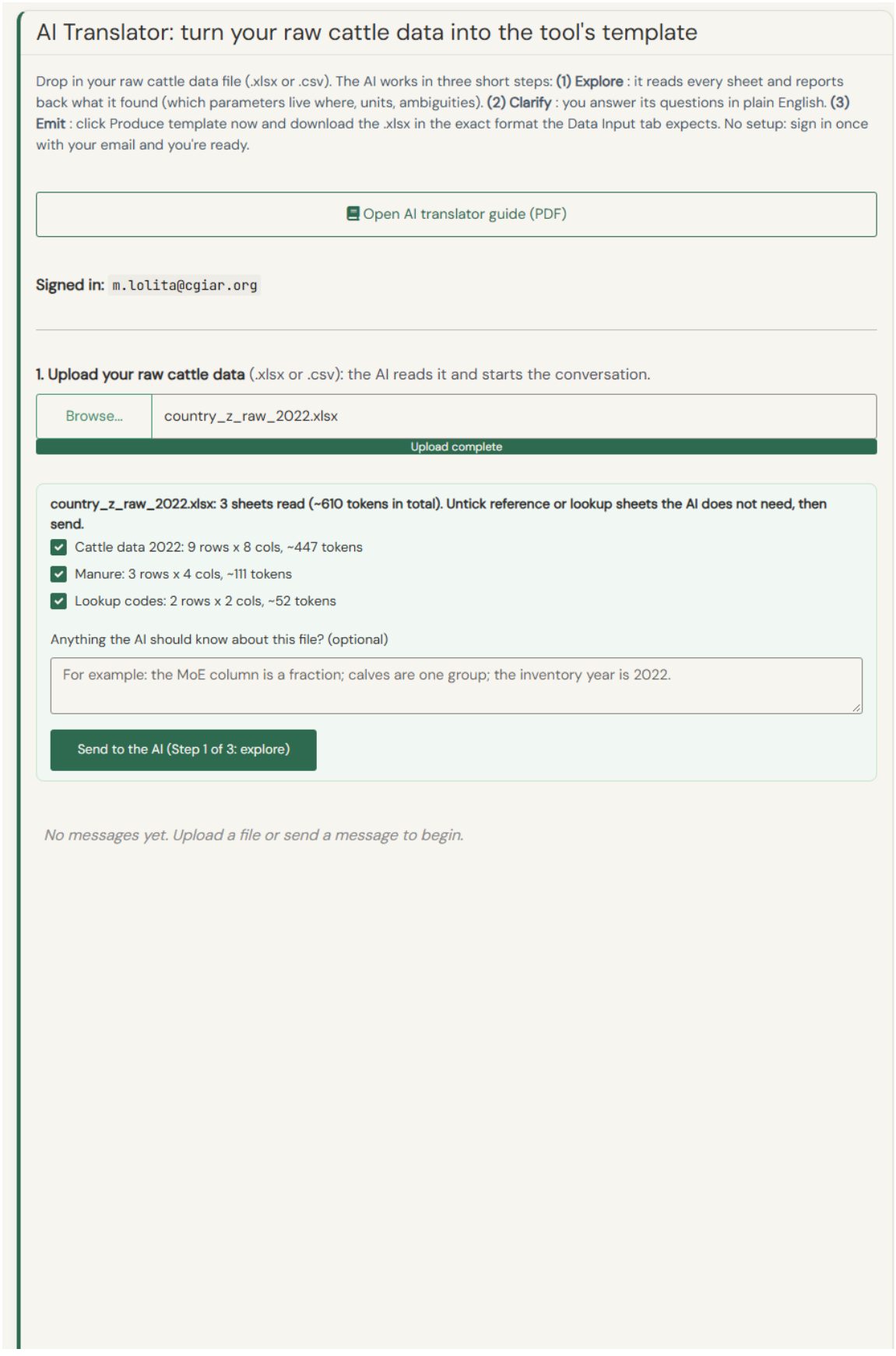
The link in the email is valid for 30 days. Once you have clicked it, the browser stays signed in.


AI Translator: turn your raw cattle data into the tool's template
Drop in your raw cattle data file (.xlsx or .csv). The AI works in three short steps: (1) Explore : it reads every sheet and reports back what it found (which parameters live where, units, ambiguities). (2) Clarify : you answer its questions in plain English. (3) Emit : click Produce template now and download the .xlsx in the exact format the Data Input tab expects. No setup: sign in once with your email and you're ready.
Open AI translator guide (PDF)
Signed in: m.lolita@cgiar.org
1. Upload your raw cattle data (.xlsx or .csv): the AI reads it and starts the conversation.
Browse...
No file selected
No messages yet. Upload a file or send a message to begin.
2. Your reply to the AI
Answer the AI's questions, or ask your own...
Send
When ready, the data workbook downloads by itself and the other two files get a button. If this page is closed before then, the three files are

2.2 2. Uploading the data

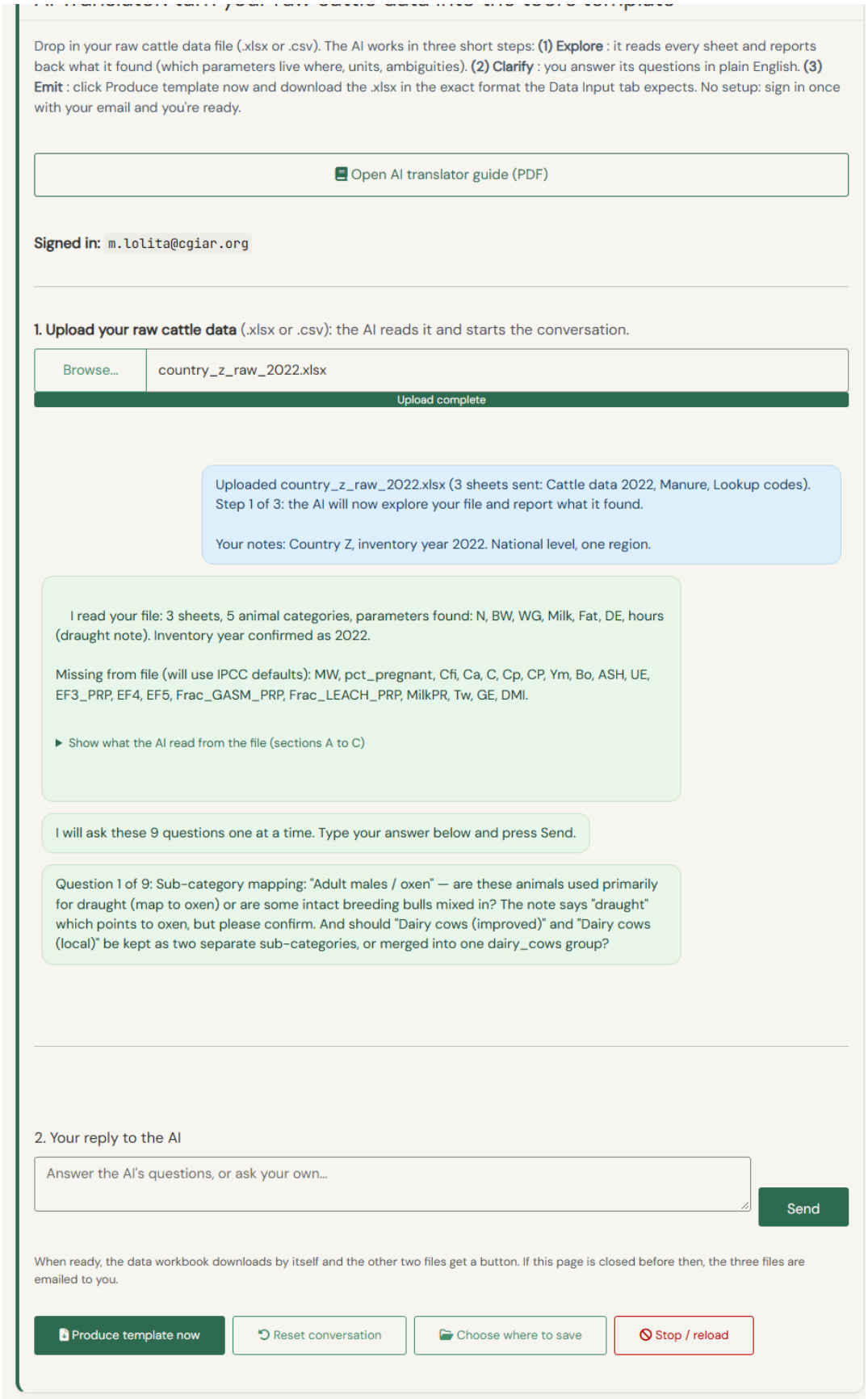
The first step is to upload your data. Click the **Browse** button and choose your inventory file. We accept Excel files, with one or several sheets, and CSV files.

You also have the option to give some context to the AI, if needed, in the box under the list of sheets.

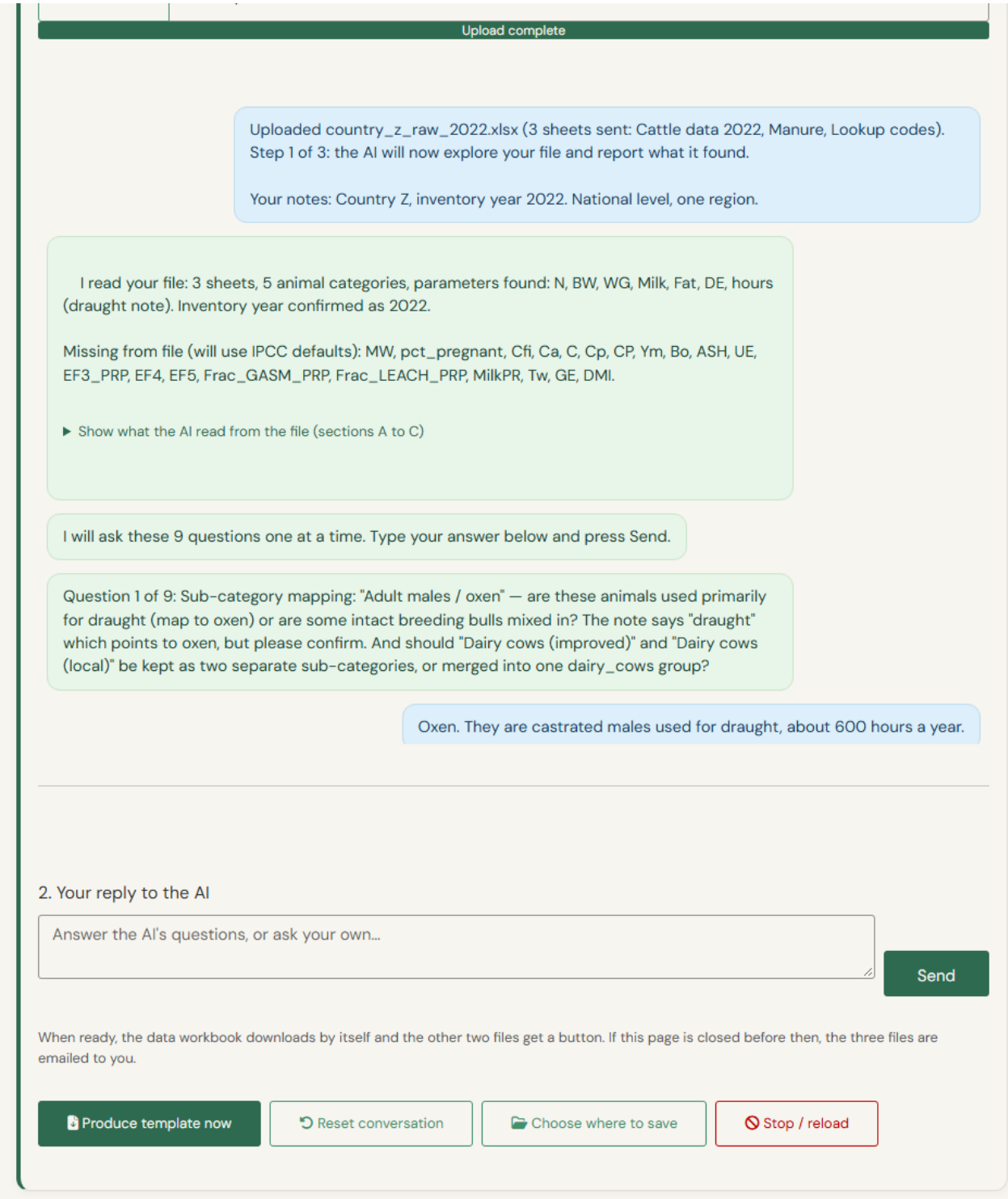

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Open AI translator guide (PDF)
Signed in: m.lolita@cgiar.org
1. Upload your raw cattle data (.xlsx or .csv): the AI reads it and starts the conversation.
Browse...
country_z_raw_2022.xlsx
Upload complete
country_z_raw_2022.xlsx: 3 sheets read (~610 tokens in total). Untick reference or lookup sheets the AI does not need, then send.
Cattle data 2022: 9 rows x 8 cols, ~447 tokens
Manure: 3 rows x 4 cols, ~111 tokens
Lookup codes: 2 rows x 2 cols, ~52 tokens
Anything the AI should know about this file? (optional)
For example: the MoE column is a fraction; calves are one group; the inventory year is 2022.
Send to the AI (Step 1 of 3: explore)
No messages yet. Upload a file or send a message to begin.

2.3 3. Data exploration

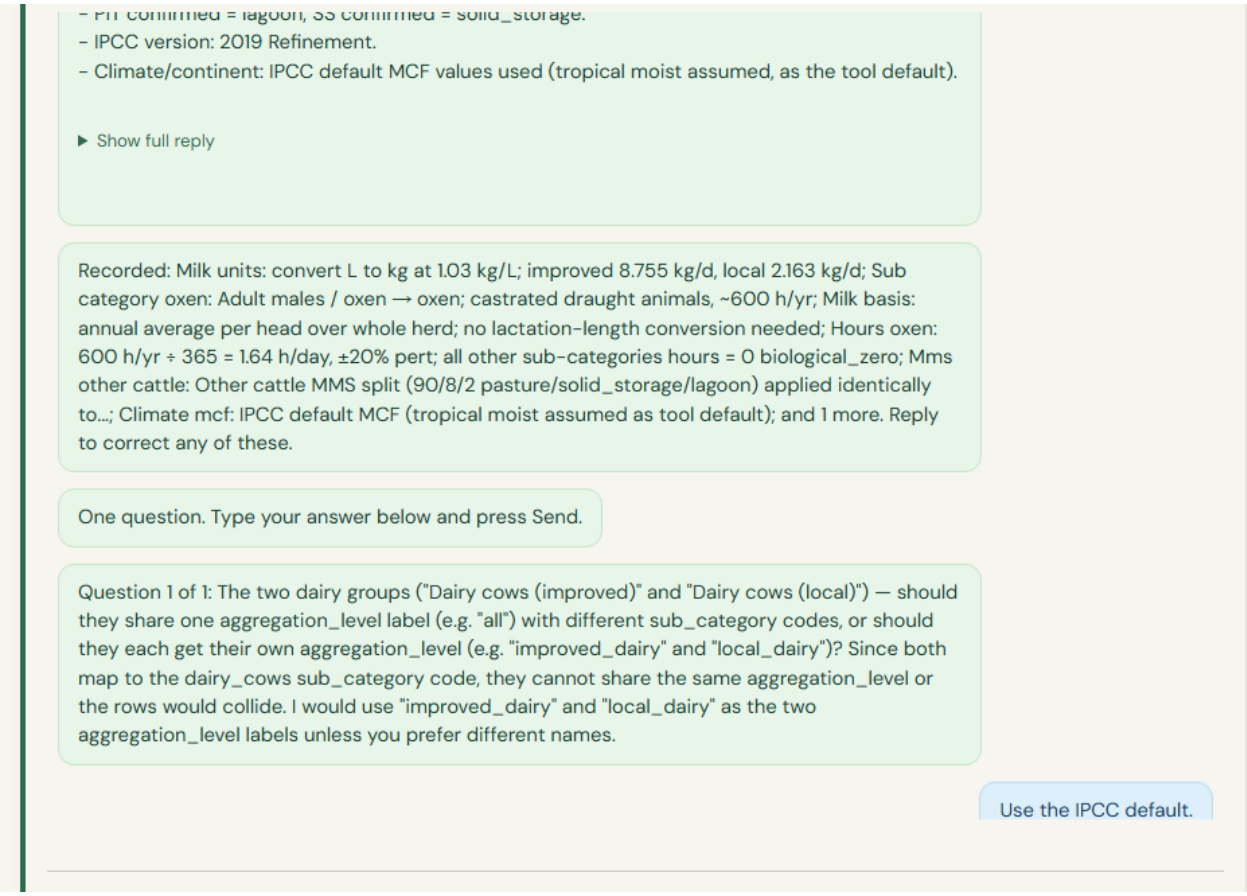
When the data is uploaded, the first thing the AI does is explore your dataset and understand it: what each label means, what the units are, and where there is an inconsistency or a possible confusion. It then returns a series of questions, to make sure it produces a high-quality dataset.


Drop in your raw cattle data file (.xlsx or .csv). The AI works in three short steps: (1) Explore : it reads every sheet and reports back what it found (which parameters live where, units, ambiguities). (2) Clarify : you answer its questions in plain English. (3) Emit : click Produce template now and download the .xlsx in the exact format the Data Input tab expects. No setup: sign in once with your email and you're ready.
Open AI translator guide (PDF)
Signed in: m.lolita@cgiar.org
1. Upload your raw cattle data (.xlsx or .csv): the AI reads it and starts the conversation.
Browse...
country_z_raw_2022.xlsx
Upload complete
Uploaded country_z_raw_2022.xlsx (3 sheets sent: Cattle data 2022, Manure, Lookup codes). Step 1 of 3: the AI will now explore your file and report what it found.
Your notes: Country Z, inventory year 2022. National level, one region.
I read your file: 3 sheets, 5 animal categories, parameters found: N, BW, WG, Milk, Fat, DE, hours (draught note). Inventory year confirmed as 2022.
Missing from file (will use IPCC defaults): MW, pct_pregnant, Cfi, Ca, C, Cp, CP, Ym, Bo, ASH, UE, EF3_PRP, EF4, EF5, Frac_GASM_PRP, Frac_LEACH_PRP, MilkPR, Tw, GE, DMI.
Show what the AI read from the file (sections A to C)
I will ask these 9 questions one at a time. Type your answer below and press Send.
Question 1 of 9: Sub-category mapping: "Adult males / oxen" — are these animals used primarily for draught (map to oxen) or are some intact breeding bulls mixed in? The note says "draught" which points to oxen, but please confirm. And should "Dairy cows (improved)" and "Dairy cows (local)" be kept as two separate sub-categories, or merged into one dairy_cows group?
2. Your reply to the AI
Answer the AI's questions, or ask your own...
Send
When ready, the data workbook downloads by itself and the other two files get a button. If this page is closed before then, the three files are emailed to you.
Produce template now
Reset conversation
Choose where to save
Stop / reload

The questions are asked one by one. You answer a question to see the next one.



Once all the questions are answered (up to 12), the answers are sent to the AI and it thinks again. It confirms what it has understood from your answers.

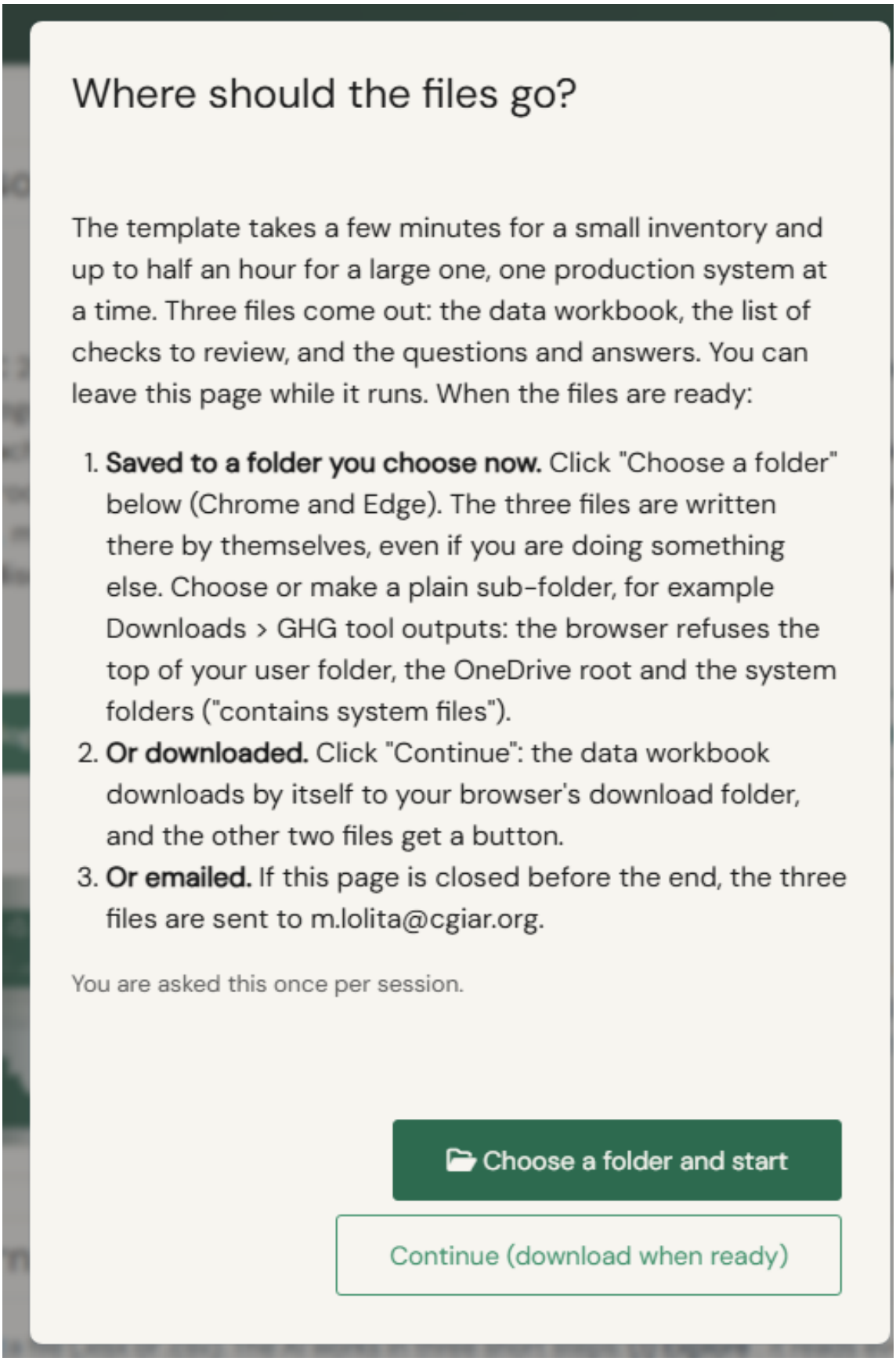


If it still has doubts, it asks other questions, and so on until everything is clear.

2.4 4. Template production

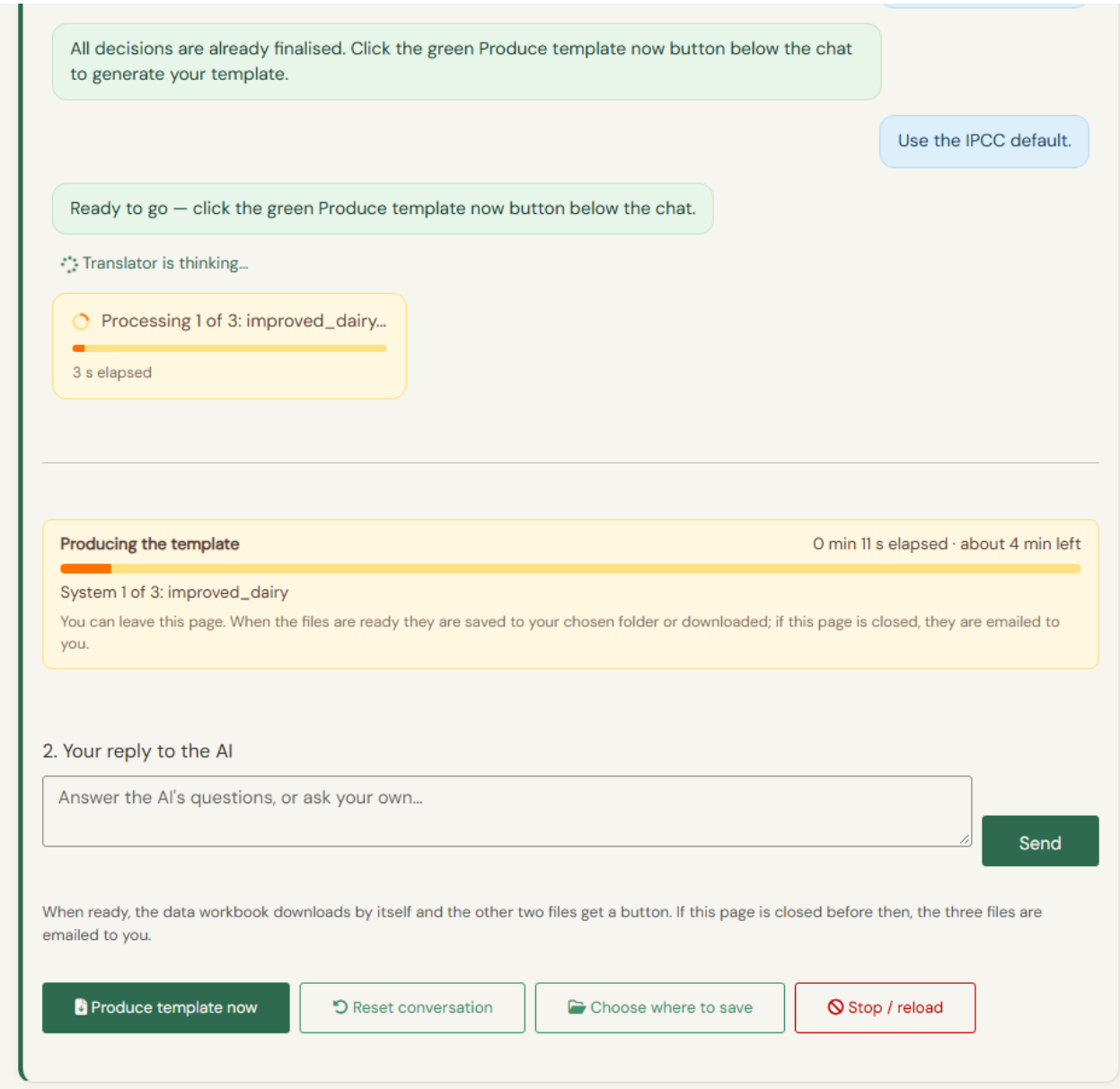
Once you are happy with the AI's answers and everything seems clear, click the **Produce template now** button.

A pop-up appears.


Where should the files go?
The template takes a few minutes for a small inventory and up to half an hour for a large one, one production system at a time. Three files come out: the data workbook, the list of checks to review, and the questions and answers. You can leave this page while it runs. When the files are ready:
1. Saved to a folder you choose now. Click "Choose a folder" below (Chrome and Edge). The three files are written there by themselves, even if you are doing something else. Choose or make a plain sub-folder, for example Downloads > GHG tool outputs: the browser refuses the top of your user folder, the OneDrive root and the system folders ("contains system files").
2. Or downloaded. Click "Continue": the data workbook downloads by itself to your browser's download folder, and the other two files get a button.
3. Or emailed. If this page is closed before the end, the three files are sent to m.lolita@cgiar.org.
You are asked this once per session.
Choose a folder and start
Continue (download when ready)

Right now, if nothing happens on the page for about 30 minutes, you are disconnected from the app. The AI is working, but that takes time, so if you are doing something else while the template is being produced, you could be disconnected and lose everything. So you have the option to authorise in advance where the outputs will go: create a folder somewhere, click **Choose a folder and start**, and give the app access to download the files there.

Then the file creation starts. You see a loading bar showing what the AI is doing, with a time estimate. This is a heavy task and can easily take up to 30 minutes.



2.5 5. Outputs

The tool produces three outputs.

Noted — no change needed, all decisions are already locked in using IPCC defaults where applicable. Everything is ready.

Click the green Produce template now button below the chat to start emission.

Recorded: Production systems kept: improved_dairy, local_dairy, non_dairy (IPCC default naming). Reply to correct any of these.

Use the IPCC default.

All decisions are already finalised. Click the green Produce template now button below the chat to generate your template.

Use the IPCC default.

Ready to go — click the green Produce template now button below the chat.

Template ready. Emitted 135 parameter rows, 15 manure_management rows, and 5 time-series rows across 3 aggregation_levels (improved_dairy, local_dairy, non_dairy). Click the green Download button below to save the .xlsx, then upload it on the 1. Data Input tab.

Your template is ready. Click the green button to save it, then open it and check the AI's work before uploading it on the 1. Data Input tab.

Quality preview of this template. 135 parameter rows, 16 manure rows.

497 pass | 1 warn | 0 fail | 34 consistency notes

Consistency notes to check against your file. These compare rows with each other. Each one is a place where the AI may have copied or invented a value.

- improved_dairy / dairy_cows / Milk: Milk = 8.755 is marked as taken from your file but no matching number is in it (2 such value(s) in all). Either the value was altered, or it was computed and should be marked user_chat rather than user_file.
- local_dairy / dairy_cows / Milk: Milk = 2.163 is marked as taken from your file but no matching number is in it (2 such value(s) in all). Either the value was altered, or it was computed and should be marked user_chat rather than user_file.
- category coverage: Source categories: 5; template groups: 5; every source category is in the template.
- Parameters vs time series: Parameters and Parameter_TimeSeries agree for the inventory year 2022 on every group and parameter checked.
- improved_dairy / dairy_cows / MCF_pct: MCF_pct came without a range; plus or minus 25% (pert) was applied as a project assumption. Replace it if your inventory has its own.
- improved_dairy / dairy_cows / EF3: EF3 came without a range; a factor of two either way (lognormal), the spread of the IPCC 2006 Table 10.21 ranges, was applied. Replace it if your inventory has its own.

Download template (.xlsx)

Download the checks (Excel) | Download the conversation (Word)

Three files: the data workbook (flagged cells are coloured, with a Check and a Why column at the right of each sheet, and filters on every column), the list of checks to review, and the questions and answers.

Back to the conversation (your answers are kept) Start a new translation

- **The translated template**, in the right format for the tool to run. Its name carries the country, the year and the time, so versions do not get mixed up.
- **The history of the conversation** with the AI, to keep a record of the questions asked and your answers.
- **The checks** you have to do on your data.

2.6 6. Checks

The AI translator is not perfect. You still have the responsibility to check your data and make sure it is correct. The tool runs some checks for you, so that you can go straight to the cells concerned. Examples of checks:

- a group of animals in your file that is missing from the template;
- a value marked as coming from your file that is not in the file;
- a percentage written as a fraction, or the reverse;
- a population counted twice, under two groups;
- manure system shares that do not add up to 100%;
- a value for which your file gave no uncertainty range, so an IPCC default range was used.

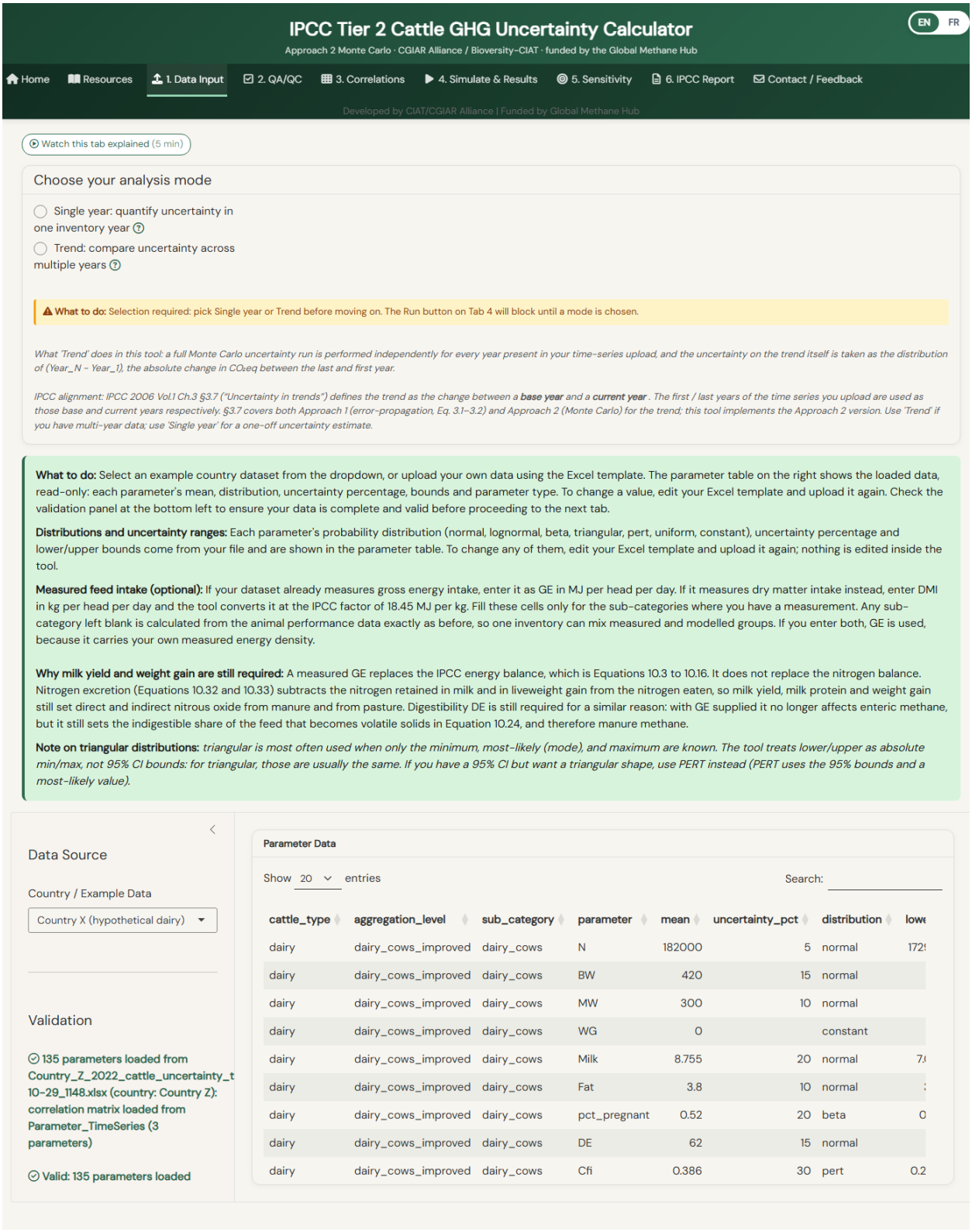
In the translated template, those cells are highlighted. Each sheet also has a **Check** column and a **Why** column at the right, with filters, so you can show only the cells that need to be checked and read the message. You can also download the list of checks as a separate file.

There are three kinds of checks:

- **Fail**: the value would break or distort the simulation. It has to be corrected before you run the tool.
- **Warn**: the value is unusual or far from the IPCC default. Check it. If it is right, keep a note of why.
- **Info**: a note for information. Nothing to correct.

2.7 7. Different routes

You can be happy with the translation and go directly to the **1. Data Input** tab to run the calculator.



Or, while checking, you realise that some things are wrong. This can come from the translator, or from the input data itself. Then you can click **Back to the conversation**, give some more information, and the AI produces the template again. You can also start a new conversation from scratch.

Good practice:

- If something is wrong in your input file, change the input file.
- If something is wrong and easy to change in the translated file, change it by hand. There is no need to run the translator again.
- If the problem is more complex, run the translator again, and explain the problem in the context box.

3 Privacy and operational cost

3.1 Where your data goes

When a file is uploaded, its contents are sent to the AI service over an encrypted connection. That is how the translation is done, and it is the main thing to be aware of before uploading anything.

A copy of the conversation, including the uploaded data and your email address, is then written to a file on the app server. It is kept so that you can close the tab and resume where you left off. It is removed in two cases: when you click **Reset conversation**, or when the server restarts after a period without use. There is no scheduled deletion beyond that, so **Reset conversation is the reliable way to remove it**.

Separately, a usage log records the date, your email address, the model and the token count of each call. It holds no file contents and exists so the administrator can monitor spending.

The AI service applies its own terms to the data it receives. Those terms are the service provider's and should be consulted directly rather than summarised here.

3.2 Cost

The translator is hosted by the GMH project on a dedicated AI service account, and you are not billed for usage. A monthly spending cap is in place. If the cap is reached, a "monthly cap reached, contact administrator" message is shown, and translation requests pause until the next cycle.

3.3 Whether to use it at all

The IPCC inventory data the translator handles is generally not sensitive: it is destined for UNFCCC reporting, which is a public process. For ministries with internal policies restricting the upload of pre-publication data to external AI services, the translator should not be used. The template can instead be filled manually, using the IPCC defaults provided in the user guide.

3.4 Related material

- User Guide, section "AI-assisted data preparation": the same workflow described in the printable user guide.
- Methodology document: full equations, parameter definitions, and references for everything the translator outputs.
- Find out more: Correlations: the companion page covering the correlation modes on the Simulate tab.